

Urban Policies to Combat Climate Change: Assessing the Impact of Congestion Pricing and Low-Emission Zones using Machine Learning Techniques

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Course: *Machine Learning in Econometrics*

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Executive Summary

Abstract content goes here. This section should provide a concise summary of the research question, methodology, key findings, and implications of the study. It should be clear and informative, allowing readers to quickly understand the purpose and results of the paper.

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Replication files are available in the [GitHub repository](#) for this project.

1 Introduction

2 Empirical Strategy and Identification

- Target parameters: ATE, ATT, GATT (possible differences in causal forest and aipw lasso estimation) (show math and explain shortly)

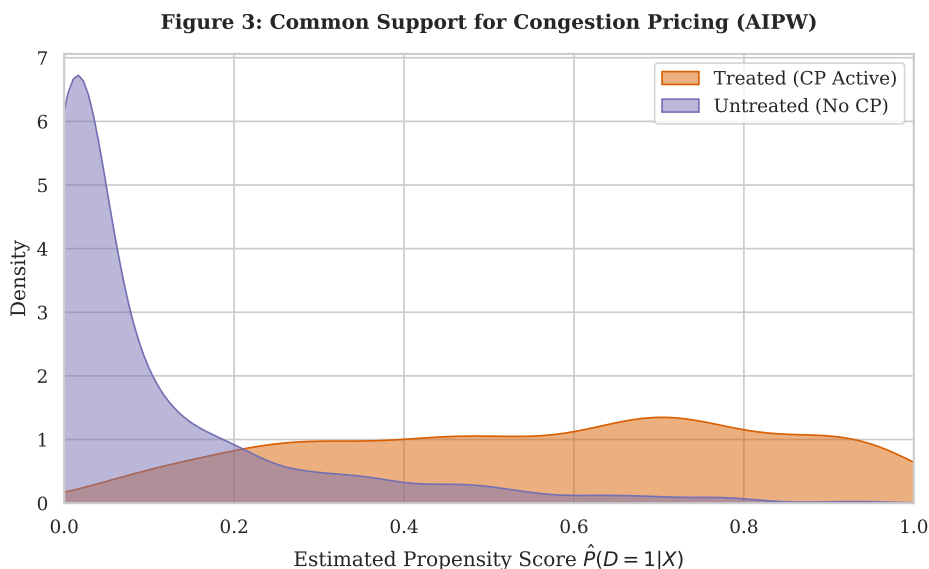


Figure 1: Common Support for Congestion Pricing (AIPW)

3 Proposed Estimator

- Use AIPW and Causal Forest instead of IPW or Reg. adj.
- Explain how they work and differences

4 Theoretical Properties

- Proof of Neyman Orthogonality (is it the same for both?)
- Not yet sure about other properties

5 Data Preparation

- no missing values or absurd numbers - check for skewed distributions (plot in appendix)
- log transform variables where it makes economic sense and where it solves skewness -
- Time invariant variables and creation of Mundlak device in order to account for panel structure (national climate pact removed because not changing)

6 K-means clustering

- determination of two clusters necessary to make heterogeneity analysis with many co-variates viable - optimal $k = 2$ (interpretability and silhouette analysis) - Cluster 1: urban metropolises, Cluster 2: sprawling regional hubs - describe them based on the cluster differences plot

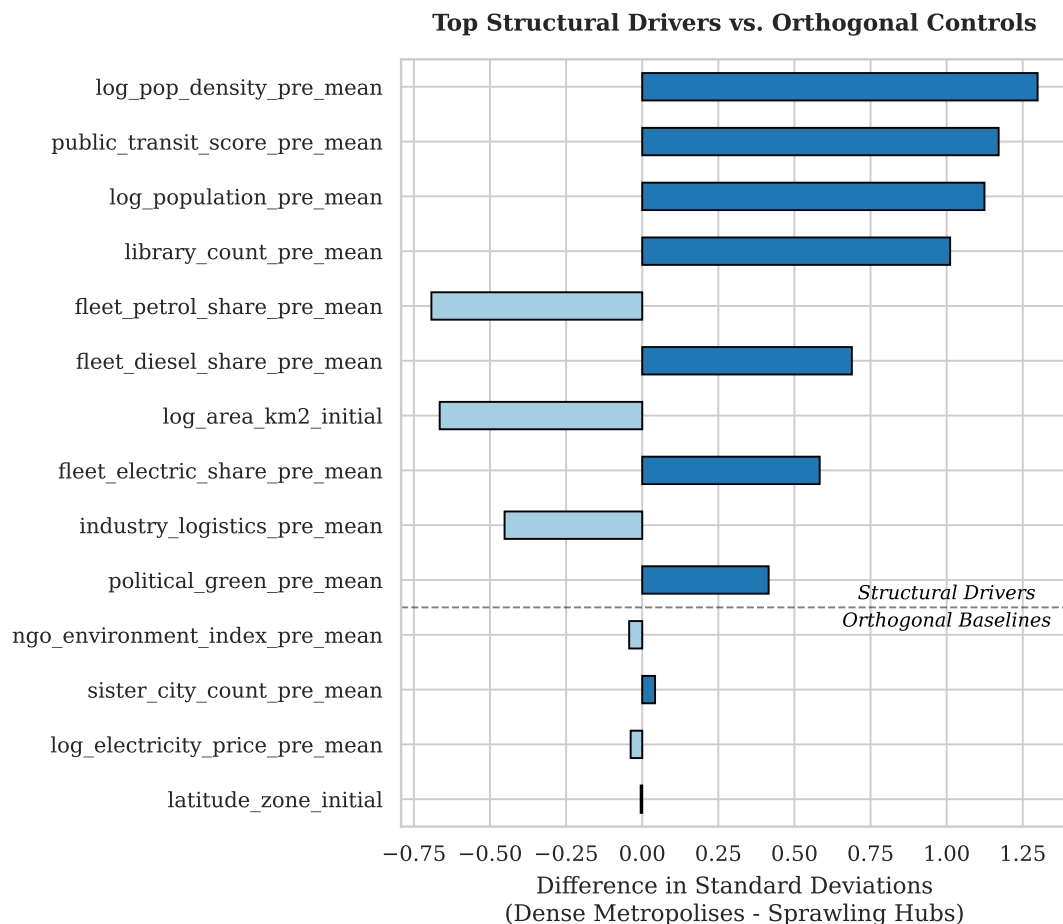


Figure 2: Differences between the two clusters.

7 Results

Notes: p -values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. 'Failed' estimation bounds denote overlap violations.

7.1 Spatial Heterogeneity

8 Sensitivity Analysis

Notes: p -values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 1: Double Machine Learning Estimates of Urban Climate Policies (Categorical Regime)

Policy	(1) OLS Baseline	(2) AIPW (Lasso)	(3) Causal Forest
Panel A: Global Average Treatment Effect (ATE)			
Congestion Pricing (CP)	–	–0.2991*** ($p = 0.002$)	–0.3829*** ($p = 0.000$)
Low Emission Zone (LEZ)	–	–0.1001 ($p = 0.157$)	–0.1528* ($p = 0.061$)
Synergy (CP $\times$ LEZ)	–	–1.0725*** ($p = 0.000$)	–0.8590*** ($p = 0.000$)
Panel B: Average Treatment Effect on the Treated (ATT)			
Congestion Pricing (CP)	–	–0.3190** ($p = 0.013$)	–0.3904*** ($p = 0.000$)
Low Emission Zone (LEZ)	–	–0.1085 ($p = 0.123$)	–0.1627* ($p = 0.056$)
Synergy (CP $\times$ LEZ)	–	–1.3667*** ($p = 0.000$)	–1.2622*** ($p = 0.000$)
Panel C: Group ATT (Heterogeneity by City Type)			
<i>Cluster 1: Dense Metropolis</i>			
Congestion Pricing (CP)	–	–0.3352* ($p = 0.061$)	–0.3966*** ($p = 0.007$)
Low Emission Zone (LEZ)	–	–0.1651* ($p = 0.089$)	–0.2299* ($p = 0.072$)
Synergy (CP $\times$ LEZ)	–	–1.4586*** ($p = 0.000$)	–1.3881*** ($p = 0.000$)
<i>Cluster 0: Sprawling Cities</i>			
Congestion Pricing (CP)	–	–0.2717*** ($p = 0.003$)	–0.3725*** ($p = 0.001$)
Low Emission Zone (LEZ)	–	–0.0508 ($p = 0.586$)	–0.0942 ($p = 0.402$)
Synergy (CP $\times$ LEZ)	–	–0.7792*** ($p = 0.002$)	–0.4570 ($p = 0.184$)

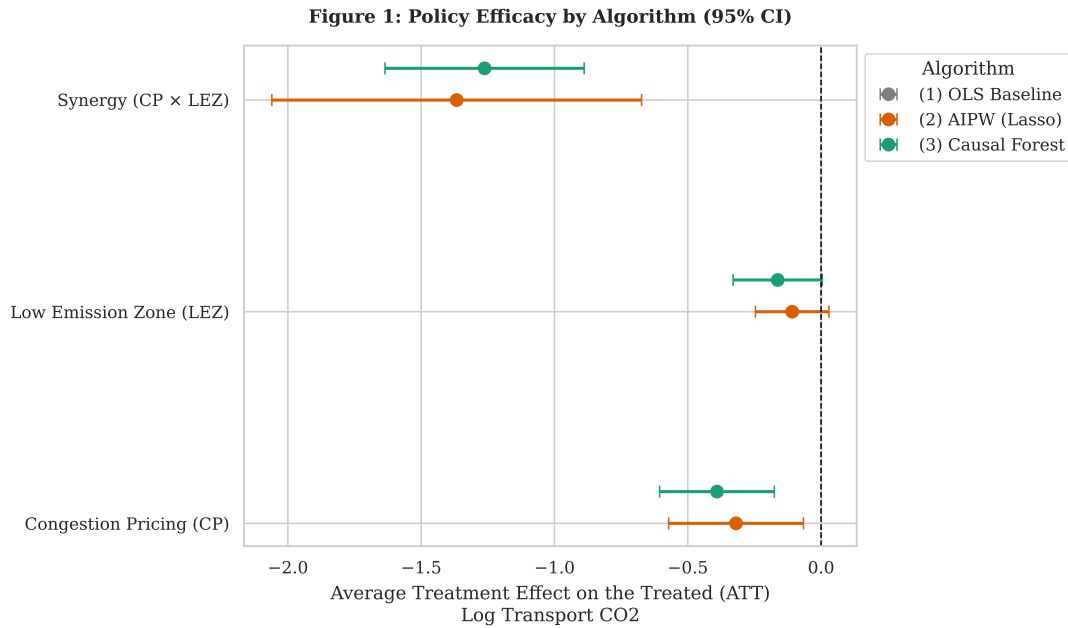


Figure 3: Policy Efficacy by Algorithm (95% CI)

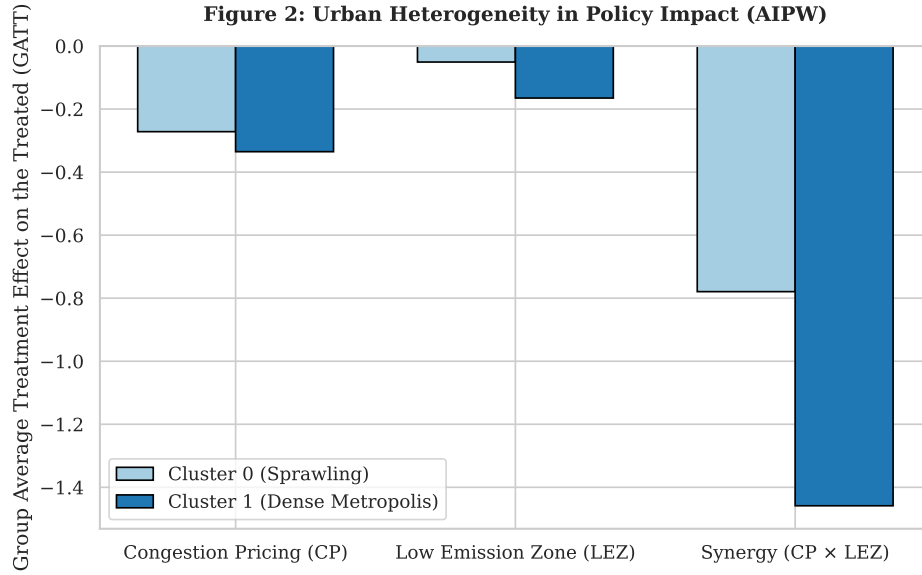


Figure 4: Urban Heterogeneity in Policy Impact (AIPW)

Table 2: Falsification Tests: Estimated Effects on Placebo Outcomes (Global ATT)

Model	AIPW (Lasso)			
	Congestion Pricing (CP)	Low Emission Zone (LEZ)	Synergy (CP × LEZ)	Control
Policy				
Decoy Outcome				
Library Count	-0.1783 ($p = 0.380$)	-0.0095 ($p = 0.916$)	0.3475 ($p = 0.334$)	-0.0001 ($p = 0.999$)

9 Interpretation and Discussion

Appendix

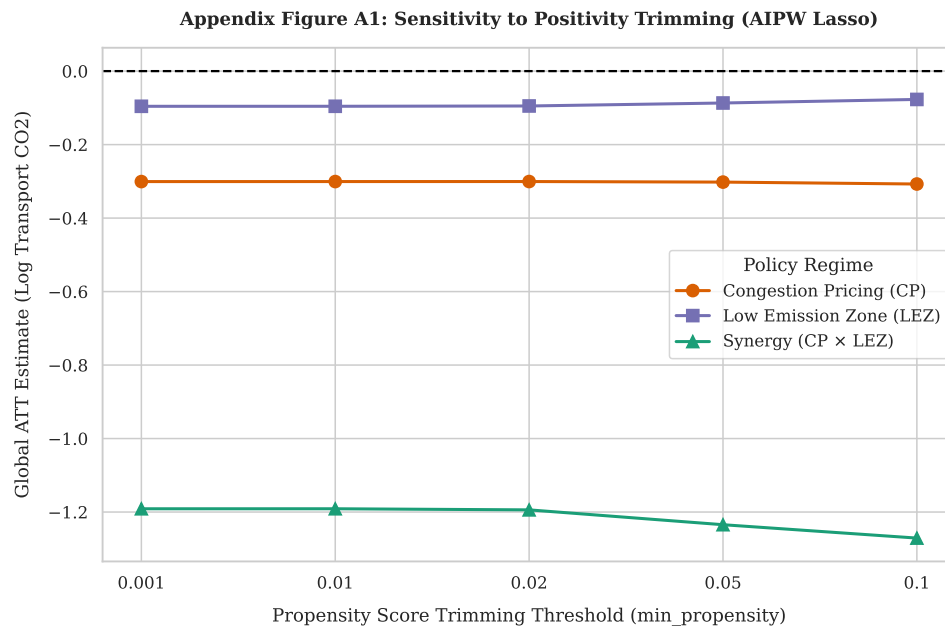


Figure 5: Sensitivity to Positivity Trimming

Declaration of Authorship

I confirm that this report is based on my own work. In preparing this report, I have not received any help from another human nor have I discussed any aspects of the empirical project with others.

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